Assignment No -4

Name :Atharv Gade **Roll No**:COB413

Title: Write a CUDA Program for : 1.Addition of two large vectors

2. Matrix Multiplication using CUDA C.

Source Code with Output :

```
!pip install cupy-cuda11x !pip
install cupy-cuda112 import
cupy as cp import numpy as
np import time


def vecAdd():     n = 1000000      h_a =
np.arange(n, dtype=np.float64)     h_b
= np.arange(n, dtype=np.float64)


    d_a = cp.array(h_a)     d_b
= cp.array(h_b)


    d_c = cp.zeros_like(d_a)


    start_time = time.time()


    d_c = d_a + d_b


    h_c = cp.asnumpy(d_c)


    end_time = time.time()


    sum_result = np.sum(h_c)     print(f"Sum: {sum_result/n}")     print(f"Time
taken for vector addition: {end_time - start_time:.6f} seconds")
```

```
    print("\nVector          A          =>\n",          h_a[:10])
print("\nVector          B          =>\n",          h_b[:10])
print("\nResultant Vector (A + B) =>\n", h_c[:10])

def matMul():     N = 4      h_a = np.random.randint(0, N,
size=(N, N), dtype=np.int32)     h_b = np.random.randint(0,
N, size=(N, N), dtype=np.int32)

    d_a = cp.array(h_a)
d_b = cp.array(h_b)

    d_c = cp.zeros((N, N), dtype=cp.int32)

    start_time = time.time()

    d_c = cp.matmul(d_a, d_b)

    h_c = cp.asnumpy(d_c)

    end_time = time.time()

    print("\nMatrix     A     =>\n",     h_a)
print("\nMatrix      B      =>\n",      h_b)
print("\nMatrix C (Result) =>\n", h_c)

    print(f"Time taken for matrix multiplication: {end_time - start_time:.6f} seconds")

def main():     print("Add two
large vectors:")     vecAdd()

    print("\nMultiply two N x N matrices:")
```

```
    matMul()


if __name__== "__main__":
main()
```

Output :

```
Add two large vectors:
Sum: 999999.0
Time taken for vector addition: 0.010821 seconds

Vector A =>
 [0. 1. 2. 3. 4. 5. 6. 7. 8. 9.]

Vector B =>
 [0. 1. 2. 3. 4. 5. 6. 7. 8. 9.]

Resultant Vector (A + B) =>
 [ 0.  2.  4.  6.  8. 10. 12. 14. 16. 18.]

Multiply two N x N matrices:

Matrix A =>
 [[0 2 3 1]
 [1 0 0 2]
 [0 0 1 1]
 [2 0 0 2]]

Matrix B =>
 [[1 0 0 0]
 [0 3 3 0]
 [2 2 0 3]
 [2 0 2 0]]

Matrix C (Result) =>
 [[ 8 13 6 9]
 [5 0 4 0]
 [4 2 2 3]
 [6 0 4 0]]

Time taken for matrix multiplication: 0.000895 seconds
```